25900323 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Core Course (DSC) - MG3DSCCSC201 - DATA STRUCTURES USING C

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Very Short Answer Questions

Answer all questions

Each question carries **2** marks

1. Define linear search. [K] / [CO1]
2. List any two basic operations of a stack. [K] / [CO2]
3. Write any two uses of a circular linked list. [K] / [CO3]
4. Define the use of base condition in recursion. [K] / [CO1]
5. What is the condition for inserting a node in the left subtree of a BST? [K] / [CO3]

[2x5 = 10]

Part B

Short Answer Questions

Answer any **5** questions

Each question carries **5** marks

6. Write the algorithm for merge sort. [K] / [CO1]
7. Convert the Infix expression A+B/C-D to prefix. [A] / [CO2]

8. Differentiate between binary tree and binary search tree in terms of structure and operations. [An] / [CO3]

9. List the difference between traversal and searching operations in data structures. [K] / [CO1]

10. Describe the array representation of a queue. How are the ‘front’ and ‘rear’ pointers managed? [U] / [CO2]

11. Compare ancestors ,descendants, and siblings of node in a tree structure. [An] / [CO3]

12. State the different parameter passing methods in functions. [K] / [CO1]

[5x5 = 25]

Part C

Essay Questions

Answer any **1** questions

Each question carries **15** marks

13. Discuss in detail the different operations on arrays with suitable examples. [U] / [CO1]

14. Explain all operations of circular queue with the help of diagram. [U] / [CO2]

[15x1 = 15]